

SEMISIMPLICITY AND THE WEIGHT FILTRATION

Let $H = (H_{\mathbb{Q}}, W, F)$ be a rational mixed Hodge structure, where $H_{\mathbb{Q}}$ is a finite-dimensional $\mathbb{Q}$ -vector space, $W = (W_w H_{\mathbb{Q}})_{w \in \mathbb{Z}}$ is its finite increasing weight filtration, and $F = (F^p H_{\mathbb{C}})_{p \in \mathbb{Z}}$ is the finite decreasing Hodge filtration on $H_{\mathbb{C}} = H_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{C}$. Thus

$$W_{w-1} H_{\mathbb{Q}} \subseteq W_w H_{\mathbb{Q}} \subseteq H_{\mathbb{Q}}, \quad F^{p+1} H_{\mathbb{C}} \subseteq F^p H_{\mathbb{C}} \subseteq H_{\mathbb{C}}.$$

Put $\mathrm{Gr}_w^W H = W_w H_{\mathbb{Q}} / W_{w-1} H_{\mathbb{Q}}$, with its induced pure Hodge structure.

Definition 1 ([D71, (2.3.4), p. 30; (2.3.8), p. 31]). An *endomorphism of the mixed Hodge structure* H is a $\mathbb{Q}$ -linear map $T : H_{\mathbb{Q}} \rightarrow H_{\mathbb{Q}}$ such that, writing $T_{\mathbb{C}} = T \otimes_{\mathbb{Q}} \mathrm{id}_{\mathbb{C}}$,

$$T(W_w H_{\mathbb{Q}}) \subseteq W_w H_{\mathbb{Q}}, \quad T_{\mathbb{C}}(F^p H_{\mathbb{C}}) \subseteq F^p H_{\mathbb{C}} \quad \text{for all } w, p \in \mathbb{Z}.$$

For such T , write $\mathrm{Gr}_w^W T$ for the induced endomorphism

$$\mathrm{Gr}_w^W T : \mathrm{Gr}_w^W H \longrightarrow \mathrm{Gr}_w^W H, \quad v + W_{w-1} H_{\mathbb{Q}} \longmapsto T(v) + W_{w-1} H_{\mathbb{Q}} \quad (v \in W_w H_{\mathbb{Q}}).$$

This is well-defined since T preserves both $W_w H_{\mathbb{Q}}$ and $W_{w-1} H_{\mathbb{Q}}$. It is a morphism of pure rational Hodge structures [D71, Theorem 2.3.5(iii), p. 30; (2.3.8), p. 31]. We denote its complexification by $(\mathrm{Gr}_w^W T)_{\mathbb{C}}$.

Semisimplicity means diagonalizability over $\mathbb{C}$, equivalently that the minimal polynomial over $\mathbb{Q}$ is square-free.

Lemma 2. *Let T be an endomorphism of a rational mixed Hodge structure H . Then T is similar to $\bigoplus_w \mathrm{Gr}_w^W T$.*

Proof. Deligne's functorial splitting [D71, Lemma 1.2.11 and (1.2.10.1), p. 13] gives a bigrading

$$H_{\mathbb{C}} = \bigoplus_{p,q} I^{p,q}, \quad W_w H_{\mathbb{C}} = \bigoplus_{p+q \leq w} I^{p,q}.$$

Since T is defined over $\mathbb{Q}$ and preserves W and F , its complexification also preserves $\overline{F}$. Functoriality therefore gives $T_{\mathbb{C}}(I^{p,q}) \subseteq I^{p,q}$ for all p, q ; see also [SS, Section 9.2, paragraph 221, p. 100]. Put $V_w = \bigoplus_{p+q=w} I^{p,q}$. Then $W_w H_{\mathbb{C}} = W_{w-1} H_{\mathbb{C}} \oplus V_w$, so the quotient projection restricts to an isomorphism

$$\phi_w : V_w \xrightarrow{\sim} (\mathrm{Gr}_w^W H)_{\mathbb{C}}, \quad \phi_w(T_{\mathbb{C}} v) = (\mathrm{Gr}_w^W T)_{\mathbb{C}} \phi_w(v).$$

Taking $\Phi = \bigoplus_w \phi_w$ under $H_{\mathbb{C}} = \bigoplus_w V_w$ shows that $T_{\mathbb{C}}$ is similar to $\bigoplus_w (\mathrm{Gr}_w^W T)_{\mathbb{C}}$. Since both operators are rational, they are similar over $\mathbb{Q}$ because they are similar over $\mathbb{C}$. $\square$

Let $f : X \rightarrow X$ be a morphism of complex algebraic varieties, and put $H^i(X) := H^i(X^{\text{an}}, \mathbb{Q})$. Deligne's [D74, Proposition 8.2.2, p. 38] gives $H^i(X)$ a canonical MHS, independent of the hypercover used to construct it, and states that pullback by every algebraic morphism is an MHS morphism. In particular,

$$f^*(W_w H^i(X)) \subseteq W_w H^i(X) \quad \text{for every } w.$$

Consequently the quotient map

$$\text{Gr}_w^W(f^*) : \text{Gr}_w^W H^i(X) \longrightarrow \text{Gr}_w^W H^i(X), \quad [\alpha] \longmapsto [f^* \alpha]$$

is well-defined: if $\alpha - \beta \in W_{w-1} H^i(X)$, then $f^* \alpha - f^* \beta \in W_{w-1} H^i(X)$. It is a morphism of pure rational Hodge structures by [D71, Theorem 2.3.5(iii), p. 30; (2.3.8), p. 31]. This is the induced action used below. No smoothness, normality, or properness assumption on X is needed.

Corollary 3. *Let $f : X \rightarrow X$ be a morphism of complex algebraic varieties. For each $i \geq 0$, $f^*|_{H^i(X^{\text{an}}, \mathbb{Q})}$ is semisimple if and only if $f^*|_{\text{Gr}_w^W H^i(X^{\text{an}}, \mathbb{Q})}$ is semisimple for every w .*

Proof. Apply Lemma 2 to the MHS endomorphism $T = f^*|_{H^i(X^{\text{an}}, \mathbb{Q})}$ just described. $\square$

Lemma 4. *Let f be a surjective endomorphism of a smooth connected projective complex variety X , and let $V \subseteq H^2(X, \mathbb{R})$ be an f^* -invariant subspace containing an ample class. If $f^*|_V$ is semisimple, then $f^*|_{H^j(X, \mathbb{Q})}$ is semisimple for every $j \geq 0$.*

Proof. Put $n = \dim X$ and $d = \deg f > 0$. By Poincaré duality and the projection formula, $f_* f^* = d \text{id}$ on each $H^j(X, \mathbb{C})$. Thus $A = f^*|_{\bigoplus_{0 \leq j \leq 2n} H^j(X, \mathbb{C})}$ is invertible. Write $A = A_s A_u$ for its multiplicative Jordan decomposition, where A_s is semisimple, A_u is unipotent, and $A_s A_u = A_u A_s$. Since A is defined over $\mathbb{Q}$, both factors commute with complex conjugation. They belong to the complex Zariski closure of the cyclic group generated by A , so they preserve the grading, Hodge decomposition and cup product; see the algebraic-group argument in [LB19, Section 1.2, Theorem 1.1, pp. 472–473]. Since $A|_{H^{2n}(X, \mathbb{C})} = d \text{id}$, its unipotent part satisfies $A_u|_{H^{2n}(X, \mathbb{C})} = \text{id}$. Hence A_u preserves integration. Put $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$. The restriction of A_u to $V_{\mathbb{C}}$ is the unipotent part of $A|_{V_{\mathbb{C}}}$, hence is the identity. Thus $A_u h = h$ for an ample class $h \in V$.

For each j , define

$$L_h : H^j(X, \mathbb{C}) \longrightarrow H^{j+2}(X, \mathbb{C}), \quad \alpha \longmapsto h \wedge \alpha.$$

Since A_u preserves cup product and $A_u h = h$, we have

$$A_u(L_h \alpha) = A_u(h \wedge \alpha) = (A_u h) \wedge (A_u \alpha) = h \wedge A_u \alpha = L_h(A_u \alpha).$$

Let $W^j = \ker(L_h^{n-j+1} : H^j(X, \mathbb{C}) \rightarrow H^{2n-j+2}(X, \mathbb{C}))$ for $0 \leq j \leq n$. For $\alpha \in W^j$,

$$L_h^{n-j+1}(A_u \alpha) = A_u(L_h^{n-j+1} \alpha) = 0,$$

so $A_u(W^j) \subseteq W^j$. On $W^j \cap H^{r,s}(X, \mathbb{C})$, where $r + s = j$, the Hodge–Riemann relations give the positive definite Hermitian form

$$(\alpha, \beta)_h = i^{r-s} (-1)^{j(j-1)/2} \int_X \alpha \wedge \bar{\beta} \wedge h^{n-j};$$

see [D71, (2.2.6)]. The operator A_u preserves this form. A unitary unipotent operator is the identity, so $A_u = 1$ on each W^j .

For $0 \leq k \leq n$, the Lefschetz decomposition [Voi10, Theorem 2.11] gives

$$H^k(X, \mathbb{C}) = \bigoplus_{0 \leq \ell \leq \lfloor k/2 \rfloor} L_h^\ell(W^{k-2\ell}).$$

Since $A_u L_h = L_h A_u$ and $A_u|_{W^{k-2\ell}} = \text{id}$, we have

$$A_u(L_h^\ell \alpha) = L_h^\ell(A_u \alpha) = L_h^\ell \alpha \quad (\alpha \in W^{k-2\ell}).$$

Thus $A|_{H^k(X, \mathbb{C})}$ is semisimple for $k \leq n$. Poincaré duality and the projection formula complete the proof. $\square$

For a projective complex variety X , let $N^1(X)_\mathbb{Q}$ be the space of Cartier divisors modulo numerical equivalence with rational coefficients, and put $N^1(X)_\mathbb{R} = N^1(X)_\mathbb{Q} \otimes_\mathbb{Q} \mathbb{R}$.

Lemma 5. *Let f be a surjective endomorphism of a projective variety X over an algebraically closed field, and let $V \subseteq N^1(X)_\mathbb{R}$ be an f^* -invariant subspace containing an ample class. If $f^*|_V$ is semisimple, then $f^*|_{N^1(X)_\mathbb{R}}$ is semisimple. Over $\mathbb{C}$, the same conclusion holds if instead $V \subseteq H^2(X, \mathbb{R})$ is an f^* -invariant subspace containing an ample class and $f^*|_V$ is semisimple.*

Proof. The assertion is immediate when $\dim X \leq 1$. Put $n = \dim X \geq 2$ and $d = \deg f$. Pullback on numerical Cartier divisor classes is injective by the projection formula, so $A = f^*|_{N^1(X)_\mathbb{R}}$ is invertible. Write $A = A_s A_u$. For $D_1, \dots, D_n \in N^1(X)_\mathbb{R}$, the projection formula gives

$$(AD_1) \cdots (AD_n) = d(D_1 \cdots D_n).$$

Thus $d^{-1/n}A$ belongs to the real algebraic group preserving the top intersection form. Its unipotent part is A_u , which belongs to the same group by [Milne17, Theorem 9.18]. Since Jordan decomposition commutes with restriction to invariant subspaces [Milne17, Proposition 9.14], semisimplicity of $A|_V$ gives $A_u|_V = \text{id}$. Choose an ample class $h \in V$. Then $A_u h = h$, so A_u preserves

$$q_h(D, E) = D \cdot E \cdot h^{n-2}, \quad h^\perp = \{D \in N^1(X)_\mathbb{R} : D \cdot h^{n-1} = 0\}.$$

By the numerical Hodge index theorem [Hu20, Proposition 2.9], $q_h|_{h^\perp}$ is negative definite, without any smoothness or normality assumption on X . Consequently $A_u|_{h^\perp}$ is orthogonal and unipotent, hence is the identity. Since $h^n > 0$, we have $N^1(X)_\mathbb{R} = \mathbb{R}h \oplus h^\perp$. Thus $A_u = \text{id}$.

For the assertion over $\mathbb{C}$, let $C(X) \subseteq H^2(X, \mathbb{R})$ be the real span of first Chern classes of line bundles. There is a natural equivariant surjection $C(X) \rightarrow N^1(X)_\mathbb{R}$: a vanishing real Chern class has degree zero on every curve. The image of $V \cap C(X)$ is an invariant subspace containing an ample class, and its action is semisimple as a subquotient of $f^*|_V$. Apply the first assertion. $\square$

Definition 6 ([D74, (5.3.5)–(5.3.8)]). Let X be a projective complex variety. An *augmented simplicial scheme* over X consists of separated finite-type complex schemes $(X_p)_{p \geq 0}$ and morphisms

$$\begin{aligned} \pi_p : X_p &\rightarrow X & (p \geq 0), \\ d_i^{(p)} : X_p &\rightarrow X_{p-1} & (p \geq 1, 0 \leq i \leq p), \\ s_i^{(p)} : X_p &\rightarrow X_{p+1} & (p \geq 0, 0 \leq i \leq p), \end{aligned}$$

called augmentation, face, and degeneracy maps, respectively. Composition is read from right to left; for example,

$$d_i^{(p-1)} \circ d_j^{(p)} : X_p \xrightarrow{d_j^{(p)}} X_{p-1} \xrightarrow{d_i^{(p-1)}} X_{p-2}.$$

The required identities are

$$\begin{aligned} d_i^{(p-1)} \circ d_j^{(p)} &= d_{j-1}^{(p-1)} \circ d_i^{(p)} \quad (p \geq 2, 0 \leq i < j \leq p), \\ s_i^{(p+1)} \circ s_j^{(p)} &= s_{j+1}^{(p+1)} \circ s_i^{(p)} \quad (p \geq 0, 0 \leq i \leq j \leq p), \\ d_i^{(p+1)} \circ s_j^{(p)} &= \begin{cases} s_{j-1}^{(p-1)} \circ d_i^{(p)}, & i < j, \\ \text{id}_{X_p}, & i = j \text{ or } i = j + 1, \\ s_j^{(p-1)} \circ d_{i-1}^{(p)}, & i > j + 1 \end{cases} \quad \begin{matrix} p \geq 0, \\ 0 \leq j \leq p, \\ 0 \leq i \leq p + 1, \end{matrix} \\ \pi_{p-1} \circ d_i^{(p)} &= \pi_p \quad (p \geq 1), \quad \pi_{p+1} \circ s_i^{(p)} = \pi_p \quad (p \geq 0). \end{aligned}$$

See [Stacks, Tag 0169]. We omit the level superscripts when the source and target are clear; thus $d_i d_j$ means $d_i^{(p-1)} \circ d_j^{(p)}$ on X_p . Write $X_\bullet = (X_p, d_i, s_i)_{p \geq 0}$ and $\pi_\bullet : X_\bullet \rightarrow X$. Define the matching schemes by

$$\begin{aligned} Y_0 &= X, \quad Y_1 = X_0 \times_X X_0, \\ Y_p &= \left\{ (x_0, \dots, x_p) \in X_{p-1}^{\times_X(p+1)} \left| \begin{array}{l} d_i^{(p-1)}(x_j) = d_{j-1}^{(p-1)}(x_i) \\ (0 \leq i < j \leq p) \end{array} \right. \right\} \quad (p \geq 2), \end{aligned}$$

with the closed subscheme structure defined by these equations. Write $\text{pr}_i : Y_p \rightarrow X_{p-1}$ for the i th projection. The matching maps are

$$\begin{aligned} \mu_0 &= \pi_0 : X_0 \rightarrow Y_0, \\ \mu_p &= (d_0^{(p)}, \dots, d_p^{(p)}) : X_p \longrightarrow Y_p, \\ x &\longmapsto (d_0^{(p)}(x), \dots, d_p^{(p)}(x)) \quad (p \geq 1). \end{aligned}$$

The face identities imply that this tuple lies in Y_p ; in particular, $\text{pr}_i \circ \mu_p = d_i^{(p)}$ for $0 \leq i \leq p$. For $p \geq 1$ and $0 \leq i \leq p$, the following diagram commutes:

$$\begin{array}{ccc} X_p & \xrightarrow{\mu_p} & Y_p \\ \pi_p \downarrow & \searrow d_i^{(p)} & \downarrow \text{pr}_i \\ X & \xleftarrow{\pi_{p-1}} & X_{p-1}. \end{array}$$

The augmentation $\pi_\bullet$ is a *proper hypercover* if every μ_p is proper and surjective. It is *smooth projective* if every X_p is a finite disjoint union of smooth projective varieties. For $f \in \text{Aut}(X)$, it is *f-equivariant* if there exist $f_p \in \text{Aut}(X_p)$ satisfying

$$\pi_p f_p = f \pi_p, \quad d_i f_p = f_{p-1} d_i, \quad s_i f_p = f_{p+1} s_i.$$

For $p, q \geq 0$, let $K^{p,q} = C_{\text{sing}}^q(X_p^{\text{an}}, \mathbb{Q})$ be the space of singular cochains, with coboundary d_{sing} . Set $K^{p,q} = 0$ if $p < 0$ or $q < 0$, and define

$$\begin{aligned} h|_{K^{p,q}} &= \sum_{i=0}^{p+1} (-1)^i (d_i^{(p+1)})^* : K^{p,q} \rightarrow K^{p+1,q}, \\ v|_{K^{p,q}} &= (-1)^p d_{\text{sing}} : K^{p,q} \rightarrow K^{p,q+1}. \end{aligned}$$

Then $h^2 = v^2 = hv + vh = 0$. Define

$$\mathrm{Tot}^n K = \bigoplus_{p+q=n} K^{p,q}, \quad D = h + v : \mathrm{Tot}^n K \rightarrow \mathrm{Tot}^{n+1} K,$$

$$H^n(X_\bullet, \mathbb{Q}) := \frac{\ker(D : \mathrm{Tot}^n K \rightarrow \mathrm{Tot}^{n+1} K)}{\mathrm{im}(D : \mathrm{Tot}^{n-1} K \rightarrow \mathrm{Tot}^n K)}.$$

Here $\bullet$ denotes the entire simplicial scheme, not an unspecified level. This cochain model computes the simplicial cohomology of [D74, (5.2.2)–(5.2.3)]. We spell out the augmentation map on singular cochains. Let $\Delta^n = \{(t_0, \dots, t_n) \in \mathbb{R}_{\geq 0}^{n+1} \mid \sum_i t_i = 1\}$. By definition,

$$C_n^{\mathrm{sing}}(X^{\mathrm{an}}, \mathbb{Q}) = \bigoplus_{\tau : \Delta^n \rightarrow X^{\mathrm{an}} \text{ continuous}} \mathbb{Q}[\tau], \quad C_{\mathrm{sing}}^n(X^{\mathrm{an}}, \mathbb{Q}) = \mathrm{Hom}_{\mathbb{Q}}(C_n^{\mathrm{sing}}(X^{\mathrm{an}}, \mathbb{Q}), \mathbb{Q}).$$

Thus a cochain assigns a rational number to each continuous singular simplex, extended linearly to finite chains. These are singular chains, not the simplicial chains of a chosen triangulation. For $c \in C_{\mathrm{sing}}^n(X^{\mathrm{an}}, \mathbb{Q})$, define

$$(\pi_0^* c)([\sigma]) = c([\pi_0^{\mathrm{an}} \circ \sigma]), \quad \sigma : \Delta^n \rightarrow X_0^{\mathrm{an}} \text{ continuous}.$$

The composite is a singular simplex in X^{an} , so this formula and linear extension define a unique element of $C_{\mathrm{sing}}^n(X_0^{\mathrm{an}}, \mathbb{Q}) = K^{0,n}$. For $n \geq 0$, the total degree is explicitly

$$\mathrm{Tot}^n K = \bigoplus_{p=0}^n C_{\mathrm{sing}}^{n-p}(X_p^{\mathrm{an}}, \mathbb{Q}).$$

Its p th component has cochain degree $n - p$. The augmentation map is

$$A^n : C_{\mathrm{sing}}^n(X^{\mathrm{an}}, \mathbb{Q}) \rightarrow \mathrm{Tot}^n K, \quad c \mapsto (\pi_0^* c, 0, \dots, 0).$$

This is a well-defined linear map: the nonzero component belongs to $K^{0,n}$, and the p th component is zero in $K^{p,n-p}$ for $p > 0$. In particular, $\pi_p^* c \in K^{p,n}$ has total degree $p + n$, so it cannot serve as the p th component of $A^n(c)$ when $p > 0$.

To verify that $A^\bullet$ is a cochain map, use the convention $(d_{\mathrm{sing}} c)(b) = c(\partial b)$. Composition with π_0^{an} commutes with the singular boundary ∂ ; hence

$$d_{\mathrm{sing}} \pi_0^* c = \pi_0^* d_{\mathrm{sing}} c.$$

On $K^{0,n}$ the vertical differential has sign $(-1)^0 = 1$. For the horizontal differential, the augmentation identities give

$$\pi_0 \circ d_0^{(1)} = \pi_1 = \pi_0 \circ d_1^{(1)},$$

so that

$$\begin{aligned} h(\pi_0^* c) &= (d_0^{(1)})^* \pi_0^* c - (d_1^{(1)})^* \pi_0^* c \\ &= \pi_1^* c - \pi_1^* c = 0. \end{aligned}$$

Consequently, in total degree $n + 1$,

$$\begin{aligned} DA^n(c) &= (d_{\mathrm{sing}} \pi_0^* c, h\pi_0^* c, 0, \dots, 0) \\ &= (\pi_0^* d_{\mathrm{sing}} c, 0, \dots, 0) = A^{n+1}(d_{\mathrm{sing}} c). \end{aligned}$$

Thus $A^\bullet$ sends cocycles to cocycles and coboundaries to coboundaries, inducing the well-defined map

$$\pi_\bullet^* = H^k(A^\bullet) : H^k(X^{\mathrm{an}}, \mathbb{Q}) \longrightarrow H^k(X_\bullet, \mathbb{Q}), \quad [c] \mapsto [(\pi_0^* c, 0, \dots, 0)].$$

This establishes the cochain map and its induced cohomology map; that the latter is an isomorphism is the cohomological descent assertion below. The compatibility with the automorphisms also holds on cochains. For every total degree $m \geq 0$, define

$$T^m := \bigoplus_{p=0}^m f_p^* : \text{Tot}^m K \longrightarrow \text{Tot}^m K,$$

$$(a_0, \dots, a_m) \longmapsto (f_0^* a_0, \dots, f_m^* a_m), \quad a_p \in K^{p, m-p}.$$

Write T for this degreewise action, so $T|_{K^{p,q}} = f_p^*$. The face identities $d_i^{(p+1)} f_{p+1} = f_p d_i^{(p+1)}$ and naturality of the singular coboundary imply

$$Th = hT, \quad Tv = vT, \quad TD = DT.$$

Moreover, $\pi_0 f_0 = f \pi_0$ gives, for every cochain c ,

$$\begin{aligned} TA^n(c) &= (f_0^* \pi_0^* c, 0, \dots, 0) \\ &= (\pi_0^* f^* c, 0, \dots, 0) = A^n(f^* c). \end{aligned}$$

Thus $TA^\bullet = A^\bullet f^*$ is an equality of cochain maps. Writing $f_\bullet^* = H^k(T)$, we obtain the commutative diagram

$$\begin{array}{ccc} H^k(X^{\text{an}}, \mathbb{Q}) & \xrightarrow{\pi_\bullet^*} & H^k(X_\bullet, \mathbb{Q}) \\ f^* \downarrow & & \downarrow f_\bullet^* \\ H^k(X^{\text{an}}, \mathbb{Q}) & \xrightarrow{\pi_\bullet^*} & H^k(X_\bullet, \mathbb{Q}). \end{array}$$

Write $d_i : X_{p+1} \rightarrow X_p$ for the face maps and, for $p, q \geq 0$, set

$$C^{p,q} = H^q(X_p, \mathbb{Q}), \quad \delta^{p,q} = \sum_{i=0}^{p+1} (-1)^i d_i^* : C^{p,q} \longrightarrow C^{p+1,q}.$$

The face identities and equivariance give

$$\delta^{p+1,q} \delta^{p,q} = 0, \quad f_{p+1}^* \delta^{p,q} = \delta^{p,q} f_p^*.$$

Set $\text{im } \delta^{-1,q} = 0$. For $p, w \geq 0$, f_p^* acts by

$$\begin{aligned} f_p^* : \frac{\ker \delta^{p,w}}{\text{im } \delta^{p-1,w}} &\longrightarrow \frac{\ker \delta^{p,w}}{\text{im } \delta^{p-1,w}}, \\ \alpha + \text{im } \delta^{p-1,w} &\longmapsto f_p^* \alpha + \text{im } \delta^{p-1,w}. \end{aligned}$$

It is well-defined since

$$f_p^*(\ker \delta^{p,w}) \subseteq \ker \delta^{p,w}, \quad f_p^*(\text{im } \delta^{p-1,w}) \subseteq \text{im } \delta^{p-1,w}.$$

Property 7 (Cohomological descent). *Let $\pi_\bullet : X_\bullet \rightarrow X$ be an f -equivariant proper smooth projective hypercover as above. Then $\delta^{p+1,q} \delta^{p,q} = 0$, these maps commute with the f_p^* -actions, and there are f^* -equivariant isomorphisms of pure Hodge structures*

$$(1) \quad \text{Gr}_w^W H^k(X, \mathbb{Q}) \simeq \frac{\ker \delta^{k-w,w}}{\text{im } \delta^{k-w-1,w}} \quad (0 \leq w \leq k),$$

where $\text{im } \delta^{-1,w} = 0$ and the other weight-graded pieces vanish. The action on the right is the induced action of f_{k-w}^* .

Proof. Proper descent [D74, (5.3.5) and (5.3.8)] gives

$$\pi_{\bullet}^* : H^k(X, \mathbb{Q}) \xrightarrow{\sim} H^k(X_{\bullet}, \mathbb{Q}).$$

Here compatibility with the degeneracy maps is used: the identities

$$d_i^{(p)} \circ f_p = f_{p-1} \circ d_i^{(p)}, \quad s_i^{(p)} \circ f_p = f_{p+1} \circ s_i^{(p)}$$

make $f_{\bullet} = (f_p)_p$ an automorphism of the simplicial scheme $X_{\bullet}$ [Stacks, Tag 0169]. Its induced action $f_{\bullet}^*$ then satisfies, by functoriality,

$$\begin{aligned} \pi_{\bullet} \circ f_{\bullet} &= f \circ \pi_{\bullet} &\implies f_{\bullet}^* \circ \pi_{\bullet}^* &= \pi_{\bullet}^* \circ f^*, \\ \pi_p \circ f_p &= f \circ \pi_p &\implies f_p^* \circ \pi_p^* &= \pi_p^* \circ f^*. \end{aligned}$$

We first introduce the filtration and its spectral sequence. For all integers s, m , put

$$L^s \text{Tot}^m K := \bigoplus_{i \geq s} K^{i, m-i} \subseteq \text{Tot}^m K, \quad D = h + v,$$

with $K^{i,j} = 0$ if $i < 0$ or $j < 0$, and $\text{Tot}^m K = 0$ if $m < 0$. This is the filtration by column degree. Explicitly, for $c = \sum_i c_i \in \text{Tot}^m K$ with $c_i \in K^{i, m-i}$,

$$c \in L^s \text{Tot}^m K \iff c_i = 0 \text{ for every } i < s.$$

We have $D(L^s \text{Tot}^m K) \subseteq L^s \text{Tot}^{m+1} K$ and $D^2 = 0$. For $m \geq 0$, this decreasing filtration satisfies $L^0 \text{Tot}^m K = \text{Tot}^m K$ and $L^{m+1} \text{Tot}^m K = 0$.

For integers $r \geq 1$ and p, q , define

$$E_r^{p,q} := \{a \in L^p \text{Tot}^{p+q} K \mid Da \in L^{p+r} \text{Tot}^{p+q+1} K\} / \sim,$$

where $a' \sim a$ means that $a' - a = u + Db$ for some u, b satisfying

$$\begin{aligned} u &\in L^{p+1} \text{Tot}^{p+q} K, & Du &\in L^{p+r} \text{Tot}^{p+q+1} K, \\ b &\in L^{p-r+1} \text{Tot}^{p+q-1} K, & Db &\in L^p \text{Tot}^{p+q} K. \end{aligned}$$

These differences form a linear subspace of the numerator: indeed, $u + Db \in L^p \text{Tot}^{p+q} K$ and $D(u + Db) = Du \in L^{p+r} \text{Tot}^{p+q+1} K$. Thus the quotient is a vector space. This is the filtered-complex construction written without auxiliary cycle or boundary notation [Stacks, Tag 012K]. Write $[a]_r$ for this equivalence class and define

$$D_r^{p,q} : E_r^{p,q} \longrightarrow E_r^{p+r, q-r+1}, \quad [a]_r \longmapsto [Da]_r.$$

The target has total degree $(p+r) + (q-r+1) = p+q+1$. Its defining conditions hold because

$$Da \in L^{p+r} \text{Tot}^{p+q+1} K, \quad D(Da) = 0 \in L^{p+2r} \text{Tot}^{p+q+2} K.$$

If $a' - a = u + Db$, then

$$\begin{aligned} Da' - Da &= Du \in L^{p+r} \text{Tot}^{p+q+1} K, \\ u &\in L^{p+1} \text{Tot}^{p+q} K = L^{(p+r)-r+1} \text{Tot}^{p+q} K. \end{aligned}$$

These are exactly the conditions making Du an allowed boundary in the target quotient. Hence $[Da']_r = [Da]_r$. Moreover, $(D_r)^2 = 0$ because $D^2 = 0$. These quotient spaces and maps form the spectral sequence of the column filtration L . Here r denotes the page number, p the column filtration degree, and $p+q$ the total cohomological degree.

For $r = 1$, taking the column- p component identifies the quotient above with vertical cohomology: the numerator condition is $va_p = 0$, the term u removes higher columns, and the column- p component of Db is vb_p . Since $v = (-1)^p d_{\text{sing}}$, this gives the canonical

identification below. The column- $(p+1)$ component of Da is $ha_p + va_{p+1}$, so the induced map is h :

$$E_1^{p,q} = H^q(X_p, \mathbb{Q}), \quad D_1^{p,q} = \delta^{p,q} = \sum_{i=0}^{p+1} (-1)^i (d_i^{(p+1)})^*.$$

Taking cohomology of this first page gives

$$E_2^{p,q} = \frac{\ker \delta^{p,q}}{\operatorname{im} \delta^{p-1,q}}.$$

The first-quadrant spectral sequence converges to the cohomology of the total complex, identified above by descent:

$$E_1^{p,q} = H^q(X_p, \mathbb{Q}) \implies H^{p+q}(\operatorname{Tot} K, D) \cong H^{p+q}(X, \mathbb{Q}).$$

Because X_p is smooth projective, $E_1^{p,q}$ is pure of weight q . The maps $\delta^{p,q}$ are morphisms of Hodge structures, so the displayed quotient $E_2^{p,q}$ is also pure of weight q .

To compute the map using h, v , let $\tilde{a}$ be the truncation of a to columns $p, \dots, p+r-1$. Then

$$\begin{aligned} a - \tilde{a} &\in L^{p+r} \operatorname{Tot}^{p+q} K \subseteq L^{p+1} \operatorname{Tot}^{p+q} K, \\ D(a - \tilde{a}) &\in L^{p+r} \operatorname{Tot}^{p+q+1} K. \end{aligned}$$

Hence $\tilde{a} \sim a$. Replace a by this truncation and write

$$a = a_0 + \dots + a_{r-1}, \quad a_j \in K^{p+j, q-j}.$$

The condition $Da \in L^{p+r} \operatorname{Tot}^{p+q+1} K$ means that the components of Da in columns $p, \dots, p+r-1$ vanish; explicitly,

$$va_0 = 0, \quad va_j = -ha_{j-1} \quad (1 \leq j < r).$$

Consequently,

$$Da = ha_{r-1} \in K^{p+r, q-r+1}, \quad D_r^{p,q}([a]_r) = [ha_{r-1}]_r.$$

If a, a' are two such truncated representatives satisfying these linear equations and having the same a_0 , then

$$a' - a \in L^{p+1} \operatorname{Tot}^{p+q} K, \quad D(a' - a) \in L^{p+r} \operatorname{Tot}^{p+q+1} K.$$

Taking $u = a' - a$ and $b = 0$ in the equivalence relation proves $[a']_r = [a]_r$ and $[Da']_r = [Da]_r$. Thus the differential is independent of the corrections. These maps are morphisms of Hodge structures [D74, Proposition (8.3.5)], and

$$E_{r+1}^{p,q} = \frac{\ker D_r^{p,q}}{\operatorname{im} D_r^{p-r, q+r-1}}.$$

Inductively, $q \neq q - r + 1$ implies $D_r^{p,q} = 0$ and $E_{r+1}^{p,q} = E_r^{p,q} = E_2^{p,q}$. We spell out the equivariance of the final identification. Since T acts separately on each column and is invertible,

$$T(L^p \operatorname{Tot}^m K) = L^p \operatorname{Tot}^m K, \quad TD = DT.$$

Consequently it induces, for every r, p, q , an automorphism

$$T_r^{p,q} : E_r^{p,q} \longrightarrow E_r^{p,q}, \quad [a]_r \longmapsto [Ta]_r.$$

Indeed, if $a \in L^p \operatorname{Tot}^{p+q} K$ and $Da \in L^{p+r} \operatorname{Tot}^{p+q+1} K$, then Ta has the same two properties because $D(Ta) = T(Da)$. If $a' - a = u + Db$ is an allowed change of representative, then

$$Ta' - Ta = Tu + D(Tb).$$

Here Tu and Tb satisfy exactly the same filtration conditions as u and b , including those on their differentials. Thus $T_r^{p,q}$ is well-defined; applying the same argument to T^{-1} proves invertibility. On representatives,

$$\begin{aligned} D_r^{p,q} T_r^{p,q}([a]_r) &= [DTa]_r = [TDa]_r \\ &= T_r^{p+r, q-r+1} D_r^{p,q}([a]_r). \end{aligned}$$

It follows that T_r preserves the kernels and images defining E_{r+1} , and its induced action there is T_{r+1} . This is also the functoriality of the filtered-complex spectral sequence [Stacks, Lemma 12.24.4]. Under $E_1^{p,q} = H^q(X_p, \mathbb{Q})$, $T_1^{p,q}$ is f_p^* . Hence, on the second page,

$$T_2^{p,q}(\alpha + \text{im } \delta^{p-1,q}) = f_p^* \alpha + \text{im } \delta^{p-1,q}, \quad \alpha \in \ker \delta^{p,q}.$$

Since all D_r vanish for $r \geq 2$, the canonical identifications $E_2^{p,q} \cong E_\infty^{p,q}$ intertwine $T_2^{p,q}$ and $T_\infty^{p,q}$.

We next describe the action on the abutment. The column filtration induces the following filtration on total cohomology:

$$\begin{aligned} L^p H^k(\text{Tot } K, D) &:= \text{im} \left(H^k(L^p \text{Tot } K, D) \longrightarrow H^k(\text{Tot } K, D) \right) \\ &= \{[a] \mid a \in L^p \text{Tot }^k K, Da = 0\}. \end{aligned}$$

The last set consists of classes admitting such a representative. Define

$$\text{Gr}_L^p H^k(\text{Tot } K, D) := \frac{L^p H^k(\text{Tot } K, D)}{L^{p+1} H^k(\text{Tot } K, D)}.$$

The preservation of the column filtration and $TD = DT$ show that $H^k(T)$ preserves each of these subspaces. For a closed representative $a \in L^p \text{Tot }^k K$, the convergence isomorphism sends its class modulo $L^{p+1} H^k$ to the stable class $[a]_\infty$. Applying T on either side gives $[Ta]_\infty$. Thus the convergence isomorphism is equivariant, as expressed by

$$\begin{array}{ccc} \text{Gr}_L^p H^k(\text{Tot } K, D) & \xrightarrow{\sim} & E_\infty^{p, k-p} \\ \text{Gr}_L^p H^k(T) \downarrow & & \downarrow T_\infty^{p, k-p} \\ \text{Gr}_L^p H^k(\text{Tot } K, D) & \xrightarrow{\sim} & E_\infty^{p, k-p}. \end{array}$$

Finally, Deligne's weight identification and its naturality give

$$\pi_\bullet^*(W_w H^k(X, \mathbb{Q})) = L^{k-w} H^k(\text{Tot } K, D)$$

[D74, (8.1.20)(i)–(ii), (8.2.1)–(8.2.2)]. In particular, W_{w-1} corresponds to L^{k-w+1} . This is the Hodge-theoretic input identifying the two filtrations; it does not follow from $TD = DT$ alone. The previously proved cochain equality $TA = Af^*$ gives

$$H^k(T) \pi_\bullet^* = \pi_\bullet^* f^*.$$

Passing to the indicated filtration quotients therefore gives an equivariant isomorphism from $\text{Gr}_w^W H^k(X, \mathbb{Q})$ to $\text{Gr}_L^{k-w} H^k(\text{Tot } K, D)$. Define $\theta_{k,w}$ to be the composite of the canonical maps

$$\begin{aligned} \text{Gr}_w^W H^k(X, \mathbb{Q}) &\xrightarrow[\sim]{\text{Gr}(\pi_\bullet^*)} \text{Gr}_L^{k-w} H^k(\text{Tot } K, D) \\ &\xrightarrow{\sim} E_\infty^{k-w, w} \xrightarrow{\sim} E_2^{k-w, w} = \frac{\ker \delta^{k-w, w}}{\text{im } \delta^{k-w-1, w}}. \end{aligned}$$

Every arrow has just been shown to intertwine the corresponding actions. On the last quotient the action is induced by f_{k-w}^* , as computed in the formula for T_2 above. Thus the required equivariance is the commutative square

$$\begin{array}{ccc} \mathrm{Gr}_w^W H^k(X, \mathbb{Q}) & \xrightarrow[\sim]{\theta_{k,w}} & \frac{\ker \delta^{k-w,w}}{\mathrm{im} \delta^{k-w-1,w}} \\ \mathrm{Gr}_w^W(f^*) \downarrow & & \downarrow f_{k-w}^* \\ \mathrm{Gr}_w^W H^k(X, \mathbb{Q}) & \xrightarrow[\sim]{\theta_{k,w}} & \frac{\ker \delta^{k-w,w}}{\mathrm{im} \delta^{k-w-1,w}}. \end{array}$$

□

Theorem 8. *Let f be an automorphism of a projective complex variety X . There is an f -equivariant proper smooth projective hypercover $\pi_\bullet : X_\bullet \rightarrow X$, with automorphisms f_p lifting f , and line bundles M_p on X_p satisfying*

$$f_p^* M_p \simeq M_p, \quad M_p \text{ is } \pi_p\text{-ample.}$$

Proof. Functorial resolution is a finite sequence of blowups with invariant centres [BM08, Theorem 1.1, p. 611]. Each blowup has an invariant tautological relatively ample bundle $\mathcal{O}(1)$. At successive stages, tensor this bundle with a sufficiently large power of the pullback of the preceding bundle. The resulting bundle is invariant and ample over the original base [Stacks, Tag 0C4K]. Start with a functorial resolution $\pi_0 : X_0 \rightarrow X$, set $U_0 = X_0$, and obtain M_0 by this construction.

Inductively, the matching scheme Y_p parametrizes compatible collections of faces from the preceding terms. In particular, $Y_1 = X_0 \times_X X_0$; for $p \geq 2$, Y_p is the closed subscheme of $X_{p-1}^{\times_X(p+1)}$ cut out by the face compatibility equations. Resolve $(Y_p)_{\mathrm{red}}$ functorially to obtain U_p and form the split term

$$X_p = U_p \sqcup X_p^{\mathrm{deg}}, \quad X_p^{\mathrm{deg}} := \coprod_{\substack{0 \leq j < p \\ \sigma : [p] \twoheadrightarrow [j]}} U_j^{(\sigma)}, \quad [p] = \{0, \dots, p\},$$

where σ is order-preserving and $U_j^{(\sigma)}$ is a copy of the scheme U_j . For each such σ , let $s_\sigma : X_j \rightarrow X_p$ be the corresponding composite of degeneracy maps. The split construction identifies the inclusion of this copy with

$$U_j^{(\sigma)} \xrightarrow{\sim} U_j \hookrightarrow X_j \xrightarrow{s_\sigma} X_p;$$

see [D74, (6.2.4)–(6.2.5), pp. 19–20] and [Stacks, Tag 0170]. Thus “degenerate” describes how these subschemes occur in the simplicial object, not their reduced scheme structure. For example, $X_1 = U_1 \sqcup U_0^{(\sigma)}$ for the unique $\sigma : [1] \twoheadrightarrow [0]$, and

$$s_0^{(0)} : X_0 = U_0 \xrightarrow{\sim} U_0^{(\sigma)} \hookrightarrow X_1, \quad d_0^{(1)} s_0^{(0)} = d_1^{(1)} s_0^{(0)} = \mathrm{id}_{X_0}.$$

The automorphism of the matching scheme lifts to U_p , and the degenerate summands inherit their automorphisms. These maps give f_p commuting with every face and degeneracy map. Denote the resolution by $\rho_p : U_p \rightarrow (Y_p)_{\mathrm{red}}$, and the inclusions by $j_p : U_p \hookrightarrow X_p$ and $\iota_p : (Y_p)_{\mathrm{red}} \hookrightarrow Y_p$. Put $a_p = (f_{p-1}^{\times_X(p+1)})|_{Y_p}$ and $g_p = f_p|_{U_p}$. For $0 \leq i \leq p$, the construction gives the commutative diagram

$$\begin{array}{ccccc}
\begin{array}{c} g_p \\ \circlearrowleft \\ U_p \end{array} & \xrightarrow{\rho_p} & \begin{array}{c} (a_p)_{\text{red}} \\ \circlearrowleft \\ (Y_p)_{\text{red}} \end{array} & \xrightarrow{\iota_p} & \begin{array}{c} a_p \\ \circlearrowleft \\ Y_p \end{array} \\
\downarrow j_p & & \searrow \mu_p & & \downarrow \text{pr}_i \\
X_p = U_p \sqcup X_p^{\text{deg}} & \xrightarrow{d_i} & X_{p-1} & \xrightarrow{\pi_{p-1}} & X. \\
\begin{array}{c} \circlearrowleft \\ f_p \end{array} & & \begin{array}{c} \circlearrowleft \\ f_{p-1} \end{array} & & \begin{array}{c} \circlearrowleft \\ f \end{array}
\end{array}$$

Each circle denotes the indicated automorphism, and every arrow is equivariant. In particular, $\mu_p j_p = \iota_p \rho_p$, $d_i = \text{pr}_i \mu_p$, and $\pi_p = \pi_{p-1} d_i$.

The external tensor product of M_{p-1} on $X_{p-1}^{\times_X (p+1)}$, restricted to $(Y_p)_{\text{red}}$, is invariant and ample over X . Combining its pullback with the invariant bundle ample relative to $U_p \rightarrow (Y_p)_{\text{red}}$ as above gives the required bundle on U_p . On the other summands use the bundles already constructed. This defines M_p and completes the induction. $\square$

Lemma 9. *Let f be an automorphism of a projective complex variety X , and let $\pi_\bullet : X_\bullet \rightarrow X$ be an f -equivariant proper smooth projective hypercover. Assume that each X_p admits a π_p -ample Cartier divisor M_p with $f_p^* M_p \sim M_p$. Suppose $f^*|_{N^1(X)_\mathbb{Q}}$ is semisimple. Then $f_p^*|_{H^*(X_p^{\text{an}}, \mathbb{Q})}$ is semisimple for every $p \geq 0$.*

Proof. Choose an ample Cartier divisor B on X . Let

$$A_p = M_p + m_p \pi_p^* B, \quad m_p \gg 0.$$

Then A_p is ample [Stacks, Tag 0892]. Since X_p is smooth projective, first Chern classes identify $N^1(X_p)_\mathbb{R}$ with a subspace of $H^2(X_p, \mathbb{R})$. Thus numerical pullback defines the subspace

$$L_p = \pi_p^* N^1(X)_\mathbb{R} + \mathbb{R}[M_p] \subseteq N^1(X_p)_\mathbb{R} \subseteq H^2(X_p, \mathbb{R}).$$

It contains $[A_p]$ and is an equivariant quotient of $N^1(X)_\mathbb{R} \oplus \mathbb{R}$, where the action on the last summand is the identity. Hence $f_p^*|_{L_p}$ is semisimple. On a disconnected X_p , take a positive power of f_p fixing every component and restrict L_p to each component. These restrictions are semisimple quotients and contain ample classes. Lemma 4 therefore applies componentwise. An invertible operator over a field of characteristic zero is semisimple if and only if any fixed positive power is semisimple. Consequently $f_p^*|_{H^q(X_p, \mathbb{Q})}$ is semisimple for every q . $\square$

Theorem 10. *Let f be an automorphism of a projective complex variety X . If $f^*|_{N^1(X)_\mathbb{Q}}$ is semisimple, then $f^*|_{H^k(X^{\text{an}}, \mathbb{Q})}$ is semisimple for every $k \geq 0$.*

Proof. Choose $\pi_\bullet : X_\bullet \rightarrow X$, f_p , and M_p as in Theorem 8. By Lemma 9, $f_p^*|_{H^q(X_p, \mathbb{Q})}$ is semisimple for all $p, q \geq 0$.

For $0 \leq w \leq k$, Property 7 identifies $f^*|_{\text{Gr}_w^W H^k(X, \mathbb{Q})}$ with the operator induced by $f_{k-w}^*|_{H^w(X_{k-w}, \mathbb{Q})}$ on an invariant subquotient. It is therefore semisimple. Lemma 2 now shows that $f^*|_{H^k(X, \mathbb{Q})}$ is semisimple for every k . $\square$

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